

of the house in real-time. Using such thermostat sensors can help monitor the temperature inside the house from anywhere using the Internet.

5. Smart door locks. Smart door locks are IoT door locks that can be controlled using a home security system and IoT OSS/app over the Internet. One can lock and unlock them remotely via the app.

6. Air-conditioners. Air-conditioning is a very important part of a house and controlling the air-conditioner remotely is a part of home automation.

7. Surveillance cameras. IoT cameras can be used for the surveillance of the house and they can give live video footage of the house remotely. Both indoor and outdoor cameras can be used to monitor the indoor and outdoor environments. Such a surveillance system can have features like motion detection and can alert whenever there is a security breach.

Apart from the above IoT devices, many other devices can be installed in a house to improve the conveni-

ence and automation. Many IoT-based home appliances like washing machines, water heaters, refrigerators, dish washers, robot floor cleaners, etc can also be installed additionally to make your house fully automated.

Controller/hub: The brain of the setup

The controller/hub used for IoT devices is the main part of the system because it controls all the IoT devices/sensors installed in a house.

The connection can either be wired or wireless, but wireless is a better option. The controller is connected to the router (gateway) of the house for Internet connectivity. The IoT controller/hub uses mainly three types of protocol standards to communicate with the IoT devices—ZigBee, Z-Wave, and Wi-Fi.

Wink and SmartThings are two most common types of IoT controllers used for home automation. With such a controller/hub, it is possible to connect hundreds of IoT devices/sensors in a house and control them with the help of a smartphone app, automatically.

Automate your homes with open source software

There are various FOSS available for home automation and the most popular ones are:

1. OpenHAB. It is a FOSS for home automation written in Java. It is used to connect and control various IoT devices installed in a house. With this app, one can perform many operations like switching on lights and fans remotely. One can use various rules to program the automation of the IoT devices and can also use voice commands to control the devices.

The project to develop this app started in 2010 and in 2013 the core functionality became an official project of the Eclipse Foundation under the name Eclipse SmartHome. OpenHAB is not only available for Android OS but also for other systems like Linux, Windows, Mac, Raspberry Pi, PINE A64, and Docker. The app can also connect to the OpenHAB cloud network



Fig. 3: Hub



Fig. 4: OpenHAB

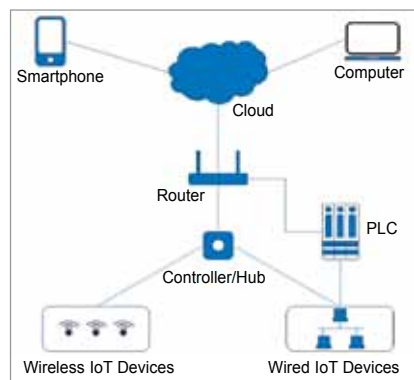


Fig. 1: Home automation setup



Fig. 2: IoT devices